

**Smart Attendance and Classroom Access Platform with
Face Verification and Geofenced Check-In
PID 12**

Software Requirements Specification

Version 1.0

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Table of Contents

1. Introduction	7
1.1 Purpose	7
1.2 Scope	7
1.3 Definitions, Acronyms, and Abbreviations	8
1.3.1 Definitions	8
1.3.2 Acronyms and Abbreviations	9
1.4 References	9
1.5 Overview	11
2. Overall Description	11
2.1 Product Perspective	11
2.2 Product Functions	11
2.3 User characteristics	12
2.3.1 Students	12
2.3.2 Lecturers	12
2.3.3 Administrators	12
2.4 Constraints	12
2.5 Assumptions and Dependencies	13
2.5.1 Assumptions	13
2.5.2 Dependencies	13
2.6 Requirements Subsets	13
2.6.1 Core Attendance Management Subset	13
2.6.2 Identity and Presence Verification Subset	13
2.6.3 Exception and Review Subset	13
2.6.4 Monitoring and Information Subset	14
2.6.5 Administration and Audit Subset	14
2.6.6 Quality and Operational Subset	14
3. Specific Requirements	14
3.1 Functionality	14
3.1.1 Authentication and User Management	14
3.1.2 Academic Information Management	15
3.1.3 Attendance Session Management	16
3.1.4 Reference Face Enrollment and Liveness Checking	17
3.1.5 Face Verification	18
3.1.6 Geofence and Dynamic QR Validation	19
3.1.7 Attendance Decision and Exception Handling	19
3.1.8 Monitoring, History, Reports, and Notifications	20
3.1.9 Audit Logging	21
3.2 Usability	21
3.2.1 Ease of Learning	21
3.2.2 Guided Attendance Process	21
3.2.3 Clear Feedback and Error Recovery	22
3.2.4 Conformance to Usability and Accessibility Standards	22

3.3 Reliability	22
3.3.1 Availability and Maintenance	22
3.3.2 Failure Recovery and MTTR	22
3.3.3 Data Integrity and Degraded Operation	23
3.3.4 Attendance and Verification Accuracy	23
3.3.5 Defect Severity and Release Quality	23
3.4 Performance and Security	24
3.4.1 Transaction Response Time	24
3.4.2 Throughput and Capacity	24
3.4.3 Degraded Operation and Resource Utilisation	24
3.4.4 Authentication and Access Security	25
3.4.5 Communication and Sensitive Data Protection	25
3.4.6 Attendance Verification Security	25
3.4.7 Auditability and Security Testing	26
3.5 Supportability	26
3.5.1 Coding Standards and Maintainable Code	26
3.5.2 Technical Documentation	26
3.5.3 Logging and Fault Diagnosis	26
3.6 Design Constraints	27
3.6.1 Architectural Constraint	27
3.6.2 Technology Stack	27
3.6.3 Standards Compliance Constraint	27
3.6.4 Client Platform and Hardware Constraint	27
3.7 On-line User Documentation and Help System Requirements	28
3.7.1 Online User Manual	28
3.7.2 Documentation Maintenance and System Information	28
3.8 Purchased Components	28
3.9 Interfaces	29
3.9.1 User Interfaces	29
3.9.2 Hardware Interfaces	36
3.9.3 Software Interfaces	36
3.9.4 Communication Interfaces	36
3.10 Database Requirements	37
3.10.1 Core Data Storage	37
3.10.2 Data Integrity and Relationships	37
3.10.3 Historical Data Preservation and Retrieval	37
3.11 Licensing, Legal, Copyright, and Other Notices	38
3.11.1 Data Protection and Privacy	38
3.11.2 Copyright, Trademarks, and Acceptable Use	38
3.12 Applicable Standards	38
3.12.1 Accessibility and User Interface Standards	38
3.12.2 Application Security Standards	38
3.12.3 Data Protection and Institutional Requirements	39

3.12.4 Date and Time Interoperability Standard	39
3.12.5 Documentation Standards	39
4. Supporting Information	39

Individual Contributions

Member	Contribution
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WITHANA W.D.P.S. 230734J	Face Verification Geofence and Dynamic QR Validation Attendance Decision and Exception Handling Monitoring, History, Reports, and Notifications Audit Logging Usability Performance and Security Communication Interfaces
WITHANAGE M.H. 230736R	Reliability Supportability Design Constraints On-line User Documentation and Help System Purchased Components Hardware Interfaces Software Interfaces Licensing, Legal, Copyright & Other Notices Applicable Standards

Software Requirements Specification

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to define the functional and non-functional requirements of the Smart Attendance and Classroom Access Platform with Face Verification and Geofenced Check-In. It describes the expected functions and behaviour of the student mobile application, lecturer and administrator web application, backend services, attendance verification components, database, and external interfaces.

The system is intended to provide secure and centralized university attendance management by allowing lecturers to manage attendance sessions and students to complete check-in using face verification, geofence validation, and dynamic Quick Response (QR) code based classroom confirmation. It also supports the management of users, courses, timetables, attendance records, reports, notifications, manual verification reviews, and audit logs.

This SRS provides a shared understanding of the system requirements among the development team, intended users, and other stakeholders. It will serve as the primary reference for system design, development, integration, testing, validation, deployment, operation, and future maintenance. It specifies what the software must provide, including its non-functional requirements, design constraints, interfaces, security needs, and other factors required for the successful development of the system.

1.2 Scope

This SRS applies to the Smart Attendance and Classroom Access Platform with Face Verification and Geofenced Check-In. The system is designed as a complete attendance management solution for universities and similar educational institutions.

The system consists of a student mobile application, a web application for lecturers and administrators, backend services, a relational database, and attendance verification components. These components work together to support attendance check-in, attendance session management, user and academic data management, attendance monitoring, reporting, notifications, and system administration.

The student mobile application allows students to complete attendance check-in and access their timetable, notifications, and attendance information. The lecturer and administrator web application supports session management, attendance monitoring, user and course management, reporting, and system configuration. The backend manages communication between the applications, attendance processing, identity and location verification, data storage, access control, and other system operations.

The system uses face verification, geofence validation, and dynamic QR based classroom confirmation to improve the reliability of the attendance records and reduce proxy attendance. It also provides a centralized platform for managing students, lecturers, courses, timetables, classrooms, attendance sessions, attendance records, and verification logs.

The system does not include full university Enterprise Resource Planning (ERP) or Learning Management System (LMS) integration, hardware biometric gates, continuous classroom surveillance, or government identity verification.

1.3 Definitions, Acronyms, and Abbreviations

1.3.1 Definitions

Term	Definition
Attendance Session	A lecturer-managed period associated with a course offering during which student attendance activities are conducted.
Initial Check-In Window	The time period at the beginning of an attendance session during which eligible students may complete face verification and geofence validation.
Late-Arrival Threshold	The configured time after which a successful initial check-in is classified as late.
QR Verification Session	An optional, time-limited verification activity activated by a lecturer during an active attendance session.
Dynamic QR Token	A short-lived, integrity-protected token displayed as a QR code and linked to a specific attendance session and QR verification session.
Face Verification	A one-to-one comparison between a live face image and the authenticated student's stored reference face embedding.
Liveness Verification	A process that checks whether the face presented to the application belongs to a live person by requiring an action such as blinking or turning the head.
Reference Face Embedding	A numerical representation of a student's facial features generated from an authorised reference image and used for face verification.
Geofence	A virtual geographical boundary defined using classroom coordinates and an allowed radius.
Geofence Validation	The process of determining whether the student's submitted location is within the geofence configured for the attendance session.
Verification Attempt	A recorded attempt by a student to complete face, liveness, geofence, or QR verification.
Attendance Record	The authoritative record containing a student's attendance result for a particular attendance session.
Pending Review	A status indicating that a verification or attendance result requires examination by an authorised lecturer or administrator.
Manual Override	An authorised change made by a lecturer or administrator to an automatically determined attendance result, together with a recorded reason.
Course Offering	A particular delivery of a course during a defined academic period, including its enrolled students and assigned lecturers.
Server-Controlled Time	Time obtained and evaluated by the backend server rather than relying on the time reported by a user's device.

Authorised User	An authenticated user who has the role and permission required to perform a particular action.
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1.3.2 Acronyms and Abbreviations

Acronym	Abbreviations
API	Application Programming Interface
ASVS	Application Security Verification Standard
CSV	Comma-Separated Values
ERP	Enterprise Resource Planning
GPS	Global Positioning System
HTTPS	Hypertext Transfer Protocol Secure
ISO	International Organization for Standardization
LMS	Learning Management System
MASVS	Mobile Application Security Verification Standard
MTTR	Mean Time to Repair
OWASP	Open Worldwide Application Security Project
PDF	Portable Document Format
QR	Quick Response
REST	Representational State Transfer
RPO	Recovery Point Objective
RTO	Recovery Time Objective
SRS	Software Requirements Specification
TLS	Transport Layer Security
WCAG	Web Content Accessibility Guidelines

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1.5 Overview

The remainder of this SRS is organised into three main sections.

Section 2, Overall Description, presents the product perspective, major system functions, user characteristics, constraints, assumptions, dependencies, and requirement subsets.

Section 3, Specific Requirements, defines the functional requirements and quality requirements, including usability, reliability, performance, security, supportability, design constraints, user documentation, interfaces, database requirements, legal requirements, and applicable standards.

Section 4, Supporting Information, contains any additional information, appendices, diagrams, or user-interface prototypes needed to support the interpretation of the requirements.

2. Overall Description

This section describes the general context, structure, users, limitations, assumptions, dependencies, and major requirement groups of the Smart Attendance and Classroom Access Platform with Face Verification and Geofenced Check-In. It provides the background needed to understand the detailed functional and non-functional requirements.

2.1 Product Perspective

The Smart Attendance Platform is a university attendance management system designed to replace or improve common attendance methods such as paper-based registers, spreadsheets, and basic digital systems. It provides a centralized platform for students, lecturers, and administrators to engage in attendance related activities.

The platform is not intended to replace the university's LMS, ERP system, student registration system, or other existing academic information systems. Instead, it uses academic information obtained from an authorised university source, either through an approved integration or through validated file imports.

The platform consists of a student mobile application, a lecturer and administrator web application, backend services, a relational database, and attendance verification components. These components communicate through secure interfaces to support attendance management and related university activities.

2.2 Product Functions

1. Authentication and access control.
2. User and academic information management.
3. Attendance session management.
4. Reference face enrollment and liveness checking.
5. Face verification.
6. Geofence and dynamic QR validation.
7. Attendance decision and exception handling.
8. Monitoring, history, reports and notification.
9. Audit logging.

2.3 User characteristics

2.3.1 Students

Students are expected to have basic experience using smartphones and mobile applications. They require a simple and guided attendance process with clear instructions for enabling permissions, completing a liveness challenge, positioning the face, improving image quality, obtaining location information, scanning a QR code, and recovering from common errors.

Students may use devices with different camera quality, processing capability, screen size, operating-system version, and network performance. The application must therefore provide understandable feedback instead of assuming that every failed verification attempt is caused by misuse.

2.3.2 Lecturers

Lecturers are expected to have basic experience using web applications, university academic systems, and classroom presentation equipment. They require only limited training to activate attendance sessions, monitor submissions, initiate additional verification activities, review failed cases, perform authorised manual updates, and generate reports. Lecturers are responsible for operating attendance sessions correctly and reviewing exceptional cases.

2.3.3 Administrators

Administrators are expected to understand the institution's user-account structure, academic periods, course information, timetable arrangements, classroom locations, and attendance policies. They manage accounts, academic-data imports, classroom and geofence configurations, system settings, reference-face review cases, reports, and audit information.

2.4 Constraints

- The system depends on ordinary mobile-device hardware. A student device must provide a usable front-facing camera, location capability, adequate processing resources, and a supported mobile operating system.
- Attendance activities are time-sensitive and may create a large number of simultaneous requests at the beginning of a lecture. Face verification is computationally more expensive than operations such as QR validation, geofence validation, or timetable retrieval. The face-verification workload must therefore be isolated and processed using controlled worker capacity so that it does not make the complete backend unresponsive.
- Face verification is probabilistic rather than perfectly accurate. Its reliability can be affected by lighting, blur, camera quality, face angle, occlusion, changes in appearance, unsuitable university reference images, and the selected comparison threshold.
- Geofence validation is limited by the accuracy of the mobile device and the availability of GPS or other location information. Indoor locations, multi-storey buildings, poor satellite visibility, and the students near the configured boundary may produce uncertain results.
- The system requires reliable internet connectivity during attendance activities, although temporary connection failures may occur.

2.5 Assumptions and Dependencies

2.5.1 Assumptions

- The university provides accurate student, lecturer, course, enrolment, timetable, and classroom information through an authorised source.
- University reference images are legally and institutionally authorised for attendance verification and are correctly associated with the relevant student IDs.
- Administrators review unsuitable, outdated, incorrectly associated, or low-quality reference images before replacement embeddings are approved.
- Students use their own authenticated accounts, supported mobile devices, and grant the camera, location, and notification permissions required for the relevant functions.
- The university provides adequate network connectivity in or near supported lecture locations. Temporary network failures may still occur and are handled through retries, pending states, or review procedures.
- Classroom coordinates and geofence radii are configured and tested by authorised Administrators before they are used in attendance sessions.
- Lecturers activate and close attendance sessions at the appropriate times and display dynamic QR codes so that they are visible only to the intended class.

2.5.2 Dependencies

- PostgreSQL, Redis, the Face Verification Service, background workers, notification services, and the deployment infrastructure are available and configured with sufficient capacity during academic operating hours.
- External notification delivery is not guaranteed to be immediate. Failure to deliver a notification does not alter an attendance session or attendance record.
- Authorised technical personnel maintain backups, monitoring, security updates, application configurations, and required third-party software dependencies.

2.6 Requirements Subsets

2.6.1 Core Attendance Management Subset

This subset covers authentication, role-based access, timetable use, attendance-session management, student eligibility, initial check-in, attendance-record creation, and finalisation. These functions form the basic operational foundation of the platform.

2.6.2 Identity and Presence Verification Subset

This subset covers university reference-image import, face-embedding enrollment, first-login trial verification, reference-face replacement, liveness challenges, image-quality validation, face preprocessing, one-to-one face verification, geofence validation, and dynamic QR verification.

2.6.3 Exception and Review Subset

This subset covers retry handling, uncertain verification results, failed submissions, pending-review cases, manual attendance corrections, excused attendance where permitted, and audit recording of all manual changes.

It provides a controlled process for legitimate cases that cannot be completed automatically.

2.6.4 Monitoring and Information Subset

This subset covers lecturer live monitoring, student attendance history, institutional attendance reports, analytics, data export, notifications, and attendance-related dashboards.

These functions use the authoritative attendance information produced by the core attendance and verification processes.

2.6.5 Administration and Audit Subset

This subset covers user administration, academic-data management, classroom and geofence configuration, system settings, reference-face reviews, institution-level reporting, and protected audit-log access.

Access is restricted to authorised administrators and technical personnel where appropriate.

2.6.6 Quality and Operational Subset

This subset covers usability, accessibility, performance, reliability, availability, data integrity, security, privacy, scalability, maintainability, logging, backup, deployment, monitoring, and technical documentation.

3. Specific Requirements

3.1 Functionality

The Smart Attendance Platform shall provide functions for users and academic data management, attendance session management, identity and classroom-presence verification, attendance decision-making, reporting, notifications, and auditing.

The main user roles are:

- Student
- Lecturer
- Administrator

The final attendance decision shall be determined by the backend using the verification methods required by the attendance policy configured for the session.

3.1.1 Authentication and User Management

FR-01: User Authentication, Account and Profile Management

The system shall allow registered students, lecturers, and administrators to log in using authorised credentials. Administrators shall be able to create, import, update, disable, reactivate, and search student and lecturer accounts, while ensuring that each account is assigned the correct role.

Each newly created or reset account shall be provided with a temporary password or secure password-setup method. Upon the first successful login, the user shall be required to create a new password that satisfies the configured password policy before accessing other protected functions.

The system shall verify the submitted credentials, account status, and assigned role before granting access to the appropriate mobile or web interface. It shall also support secure logout, password recovery, password change, and automatic session expiration.

Users shall be able to view their profile information and update only the fields permitted by institutional policy. Institution-controlled information, including student IDs, assigned roles, course enrolments, and lecturer assignments, shall be modified only by authorised personnel.

Disabling an account shall prevent future access without deleting its existing attendance records, verification history, or audit information.

FR-02: Role-Based and Platform-Specific Access Control

The system shall restrict access to functions and information according to the authenticated user's role and the application provided for that role.

Students shall use the mobile application to view their timetable, receive notifications, complete attendance verification activities, and view their attendance history. Lecturers shall use the web dashboard to manage assigned attendance sessions, monitor live attendance, review verification failures, perform authorised manual updates, and generate reports. Administrators shall use the web dashboard to manage accounts, academic data synchronisation, classrooms, geofences, system settings, reports, and audit information.

The backend shall verify the user's permissions before processing every protected request. Users shall not gain access to functions assigned to another role by entering a web address manually, modifying a request, or using a different client application.

3.1.2 Academic Information Management

FR-03: Academic Data Import and Synchronisation

The system shall obtain course offerings, academic periods, student groups, enrolled Students, and assigned lecturers from the university LMS or another authorised academic information source.

The system shall support synchronisation through an approved integration or import through a validated structured file when direct integration is unavailable. It shall not replace the LMS as the primary system for module registration or course enrolment.

The attendance platform shall maintain only the academic information needed to create attendance sessions, identify eligible students and lecturers, and produce attendance reports. Synchronisation updates shall not remove historical attendance records.

FR-04: Attendance Eligibility and Lecturer Assignment

The system shall use imported or synchronised enrolment and lecturer-assignment information to control access to attendance functions.

Only students enrolled in a course offering shall be eligible to participate in its attendance sessions. Only lecturers assigned to that course offering shall be allowed to create, manage, review, and finalise its attendance sessions unless additional permission is granted by an administrator.

Changes to current enrolments or lecturer assignments shall affect future attendance activities but shall not alter records created for completed sessions.

FR-05: Timetable Management

The system shall allow authorised administrators or lecturers to create, update, cancel, and view timetable entries required for attendance management.

Each timetable entry shall include the course offering, lecturer, student group, classroom, date, start time, and end time. The system shall identify scheduling conflicts involving the same lecturer, classroom, or student group.

Approved timetable changes shall be reflected in the relevant student and lecturer interfaces without removing historical information connected to completed attendance sessions.

FR-06: Classroom and Geofence Management

The system shall allow administrators to create and manage classroom or lecture-location information.

Each location shall include its building information, room identifier, geographical coordinates, permitted geofence radius, and operational status. The configured classroom and geofence shall be associated with the relevant timetable entries and attendance sessions.

Changes to a classroom or geofence configuration shall not modify attendance decisions already recorded using an earlier configuration.

3.1.3 Attendance Session Management

FR-07: Create, Activate, and Manage Attendance Session

The system shall allow an assigned lecturer to create an attendance session for a synchronised course offering. Each session shall include the course, timetable entry, classroom, scheduled time, eligible students, initial check-in window, late-arrival threshold, and enabled verification methods.

The system shall prevent lecturers from creating attendance sessions for course offerings to which they are not assigned. A newly created session shall remain in a draft or scheduled state until it is activated.

The assigned lecturer shall be able to activate the session and control its attendance activities. Attendance submissions shall be accepted only while the main session and the relevant verification windows are active.

During an active session, the lecturer shall be able to view the session status, eligible students, completed submissions, failed attempts, and pending results. The system shall prevent unauthorized users from activating, modifying, or controlling the session.

FR-08: Initial Attendance Check-In Window

The system shall open an initial attendance check-in window at the beginning of every attendance session. During this window, eligible students shall complete face verification and geofence validation.

The time at which the student successfully completes the required checks shall be recorded as the check-in time. The system shall classify the student as on time or late according to the configured late-arrival threshold. A student who completes the required checks after the on-time period but before the initial check-in window closes shall be recorded as late.

When QR verification is not enabled for the session, attendance may be determined using the face verification result, geofence validation result, and recorded check-in time.

FR-09: Manage Dynamic QR Verification Sessions

The system shall allow the assigned lecturer to create and activate any number of time-limited dynamic QR verification sessions at any point during an active attendance session. QR verification shall be optional, and the lecturer may conduct one, several, or no QR verification sessions during a lecture.

For each QR verification session, the lecturer interface shall display the changing QR code, remaining verification time, and number of student submissions. The lecturer shall also be able to close or cancel an active QR verification session.

Each QR verification session and its student results shall be recorded separately and linked to the main attendance session.

FR-10: Close, Cancel, and Finalise Attendance Session

The system shall allow the assigned lecturer or an authorised administrator to close, cancel, or finalise an attendance session.

Closing a session shall prevent new submissions while allowing pending and failed cases to be reviewed. Cancelling a session shall require a reason and shall preserve the cancellation record.

Finalising a session shall confirm its attendance results. Changes made after finalisation shall require an authorised correction process and shall be recorded in the audit log.

3.1.4 Reference Face Enrollment and Liveness Checking

FR-11: University Reference Image Import and Face Enrollment

The system shall obtain authorised student reference images from the university database and use them to generate the initial reference face embeddings required for one-to-one verification.

Before generating an embedding, the system shall validate that the image is associated with the correct student, contains one visible face, and has sufficient quality for face verification. The generated embedding shall be securely linked to the student ID together with its enrollment date, and status.

The original university image and generated embedding shall be accessible only to authorised personnel and services.

FR-12: First-Login Trial Verification and Reference Face Update

After the student's first login and mandatory password change, the system shall provide a trial face verification process using the reference embedding generated from the university image.

The trial shall allow the student to confirm whether face verification works successfully before participating in a real attendance session. The trial result shall not create or modify an attendance record.

If the trial repeatedly fails because the university image is outdated, unsuitable, or no longer represents the student accurately, the system shall instruct the student to contact an administrator and shall allow the account to be flagged for reference-face review.

The administrator shall be able to capture or approve a new student photograph under suitable conditions, validate its quality, generate a replacement face embedding, and associate it with the correct student ID. The replacement and the responsible administrator shall be recorded for

auditability.

FR-13: Active Liveness Verification

The mobile application shall present a randomly selected liveness challenge before accepting a face image for verification.

The challenge may require the student to blink, turn the head, or perform another supported action. The application shall use the live camera and face-detection information to determine whether the action is completed within the permitted time.

The system shall reject failed challenges, captures containing multiple faces, and attempts to use an existing gallery image. A configured number of retries shall be provided.

FR-14: Live Face Capture and Image Quality Validation

After successful liveness verification, the mobile application shall capture a live face image and submit one suitable frame to the backend.

The system shall validate the image for face visibility, brightness, blur, face position, head orientation, image resolution, and the presence of multiple faces.

The system shall reject images affected by poor lighting, excessive blur, incorrect face angles, or multiple visible faces and shall provide clear guidance that allows the student to retry.

3.1.5 Face Verification

FR-15: Face Image Preprocessing

The Face Verification Service shall prepare the submitted live face image before extracting its face embedding.

The preprocessing shall include face detection, cropping, resizing, alignment, and normalisation according to the requirements of the selected model.

If the image cannot be processed reliably, the service shall return an appropriate failure result instead of continuing with an unreliable comparison.

FR-16: One-to-One Face Verification

The Face Verification Service shall generate an embedding from the submitted live image and compare it only with the authenticated student's active reference embedding.

The comparison shall use the approved similarity method and configured verification threshold. The system shall perform one-to-one verification and shall not search the submitted face against all registered Students.

The initial implementation shall use the DeepFace with a pretrained ArcFace model and cosine-based comparison [17].

FR-17: Face Verification Result Management

The Face Verification Service shall return the match decision, comparison score or distance, image-quality result, timestamp, and failure reason where applicable.

The result shall be passed to the attendance verification component and stored with the related verification attempt. A timeout, unavailable service, invalid image, missing reference embedding, or processing error shall not be treated as a successful match.

The system shall record the threshold versions used so that verification decisions remain traceable.

3.1.6 Geofence and Dynamic QR Validation

FR-18: Geofence Validation

During an attendance verification activity, the mobile application shall obtain the student's current coordinates, reported location accuracy, and capture time.

The backend shall compare the submitted information with the geofence configured for the attendance session. Missing, stale, invalid, or clearly out-of-range locations shall be rejected. Uncertain results caused by poor location accuracy or proximity to the geofence boundary may be marked for lecturer review.

The system shall collect location data only during attendance verification activities and shall not continuously track students.

FR-19: Dynamic QR Generation, Scanning, and Validation

For each lecturer-activated QR verification session, the system shall generate a short-lived, integrity-protected QR token linked to both the main attendance session and the individual QR verification session.

The QR code shall change at the configured interval and remain valid only during its active verification window. The student mobile application shall allow an eligible student to scan and submit the code.

The backend shall validate the student's authentication and enrolment, QR integrity, attendance session, QR verification session, issue time, expiry time, and previous submission state using server-controlled time.

Expired, altered, duplicated, reused, or session-mismatched tokens shall be rejected. Successful completion shall be recorded against the particular QR verification session.

3.1.7 Attendance Decision and Exception Handling

FR-20: Attendance Verification Decision

The Attendance Verification component shall determine each student's final attendance result using the verification evidence collected during the lecture.

The initial face verification, geofence validation, and recorded check-in time shall form the basic attendance evidence. When the lecturer activates one or more QR verification sessions, the system shall also include the student's QR verification results together with the enrolment, and session-timing information.

The configured attendance policy shall determine how the collected evidence affects the final result. When no QR verification is activated, the absence of QR verification results shall not cause attendance to fail.

The final decision shall be determined by the backend rather than by the mobile application.

FR-21: Attendance Record and Status Management

The system shall maintain one authoritative attendance record for each student and attendance session.

The record shall include the initial check-in time, on-time or late classification, face verification result, results of the QR checks, final attendance status, decision source, and decision timestamp.

Supported statuses shall include Pending, Present, Late, Absent, Verification Failed, Pending Review, Manually Approved, Manually Rejected, and Excused where permitted by institutional policy.

Repeated submissions shall update the existing verification process without creating duplicate final attendance records.

FR-22: Retry and Manual Override

The system shall allow controlled retries when a verification attempt fails because of a recoverable problem such as poor image quality, liveness failure, temporary network interruption, uncertain GPS accuracy, an expired QR token, or a temporary processing error.

When a legitimate case cannot be completed successfully, an authorised lecturer shall be able to update the attendance result manually after providing a reason.

The system shall preserve the original verification evidence and record the previous status, updated status, lecturer, student, session, reason, and timestamp.

3.1.8 Monitoring, History, Reports, and Notifications

FR-23: Live Attendance Monitoring

The system shall allow the lecturer to monitor attendance progress during an active session.

The dashboard shall display eligible students, initial check-in results, late arrivals, successful and failed verification attempts, pending results, review-required cases, and students who have not responded.

The dashboard shall show suitable failure categories without exposing raw face embeddings, verification thresholds, QR secrets, or unnecessary biometric details.

FR-24: Attendance History

The system shall allow students to view their personal attendance history by course, date, academic period, and attendance status.

Lecturers shall be able to view attendance information for assigned courses, while administrators shall access authorised institution-level attendance information.

The history shall show the recorded check-in time, on-time or late classification, final status, and whether the result was automatically determined or manually updated.

FR-25: Reports and Analytics

The system shall generate attendance reports and analytics by student, attendance session, course

offering, student group, department, date range, academic period, and attendance status.

Authorised users shall be able to view attendance percentages, absence counts, late arrivals, attendance trends, students below a configured threshold, and manual override summaries.

Reports shall be exportable in approved formats such as CSV or PDF, subject to role-based access permissions.

FR-26: Attendance Notifications

The system shall notify affected users about upcoming classes, initial check-in windows, active QR verification sessions, attendance results, timetable changes, manual attendance updates, and review outcomes.

Failure to deliver a notification shall not change the related attendance session, verification result, or attendance record.

3.1.9 Audit Logging

FR-27: System Audit Logging

The system shall record important authentication, user-management, academic-data synchronisation, attendance-session, biometric enrollment, verification, manual-override, reporting, and configuration activities.

Each audit record shall contain the responsible user or service, action performed, affected entity, server-generated timestamp, and outcome. Where applicable, it shall also record the previous and updated values and a request identifier.

Audit records shall support accountability, dispute resolution, security review, and investigation of attendance changes. Students and ordinary lecturers shall not be permitted to modify or delete them.

3.2 Usability

The system shall provide simple, consistent, and role-appropriate interfaces for students, lecturers, and administrators. Users shall be able to complete common attendance tasks with minimal training, clear guidance, and understandable feedback.

3.2.1 Ease of Learning

A student with basic mobile application experience shall be able to use the application after no more than 10 minutes of orientation. The student shall be able to log in, complete the trial face verification, complete attendance check-in, and view the attendance result without technical assistance.

A lecturer shall be able to perform the main attendance-management tasks after no more than 30 minutes of training. These tasks shall include activating an attendance session, initiating QR verification sessions, monitoring attendance, reviewing failed attempts, and finalising attendance records.

3.2.2 Guided Attendance Process

The student mobile application shall guide the student through each required attendance step, including liveness verification, face capture, location validation, and QR scanning when activated by

the lecturer.

The application shall clearly indicate which activities are completed, pending, failed, expired, or awaiting lecturer review. Instructions shall be brief, understandable, and displayed at the point where the student must perform the action.

3.2.3 Clear Feedback and Error Recovery

The system shall provide clear feedback after important actions, including login, password change, face verification, QR submission, location validation, manual attendance updates, and session finalisation.

When an action fails, the system shall explain the problem in plain language and provide a suitable corrective action. For example, it shall inform the student when the image is too dark, the face is not positioned correctly, location permission is disabled, the QR code has expired, or the network connection has failed.

3.2.4 Conformance to Usability and Accessibility Standards

The web dashboard and online user manual shall conform to WCAG 2.2 Level AA requirements where applicable [1]. The student mobile application shall follow relevant Material Design 3 [7] and Apple Human Interface Guidelines [8] to provide familiar and consistent behaviour across supported Android and iOS devices.

The mobile and web interfaces shall provide readable text, sufficient colour contrast, clear navigation, accessible labels for controls, suitable touch-target sizes, keyboard accessibility for web functions, and understandable form-validation messages. Important information, including attendance and verification status, shall not be communicated using colour alone [1], [7], [8].

3.3 Reliability

The system shall remain dependable during scheduled lecture periods and shall protect attendance records from duplication, loss, or incorrect updates. Since attendance marking is time-sensitive, failures in verification, network communication, or supporting services shall be handled without falsely confirming attendance or corrupting stored information.

3.3.1 Availability and Maintenance

The production system shall target at least 99.5% availability during configured academic operating hours, excluding approved maintenance periods.

Planned maintenance shall normally be performed outside scheduled lecture periods. Administrators shall be informed in advance when maintenance is expected to affect attendance functions.

The availability target shall cover the mobile and web-facing backend services required for login, attendance-session management, verification submission, attendance decisions, and record retrieval.

3.3.2 Failure Recovery and MTTR

The target MTTR for a critical system failure shall be no more than four hours after the failure has been detected.

Critical attendance services shall be monitored so that failures in the backend, database, Face Verification Service, Redis service, or supporting workers can be identified promptly.

Permanent data shall be backed up according to the approved backup schedule. The system shall target a RTO of four hours and a RPO of no more than 24 hours for permanent attendance and administrative data.

3.3.3 Data Integrity and Degraded Operation

The system shall maintain one authoritative attendance record for each student and attendance session. Repeated requests, duplicate QR submissions, network retries, or repeated mobile actions shall not create duplicate final records.

When a non-critical service such as notification delivery, report generation, or analytics is unavailable, core attendance marking and record storage shall continue where technically possible.

When the Face Verification Service is temporarily unavailable, the system shall not record the student as successfully verified. The attempt shall be rejected, queued for controlled retry, or marked for review according to the failure condition.

When the primary database is unavailable, the system shall not display an attendance submission as successful unless the result has been stored safely. The user shall receive an appropriate failure or pending message.

3.3.4 Attendance and Verification Accuracy

Deterministic functions such as QR expiry validation, duplicate-submission detection, late-status calculation, course-enrolment validation, and attendance-percentage calculation shall produce the correct result for all approved functional test cases.

The face-verification threshold shall be evaluated using a representative test dataset containing enrolled students under supported capture conditions. The selected threshold shall target:

- A false acceptance rate of no more than **1%**
- A genuine verification success rate of at least **95%**

Verification attempts that do not meet the configured threshold shall not be recorded as successful matches. Genuine failures caused by image quality, outdated reference images, or device conditions shall support retry or authorised review.

These values are acceptance targets and must be confirmed during face-verification testing rather than assumed from the pretrained model.

3.3.5 Defect Severity and Release Quality

Defects shall be classified as follows:

Critical defect: A failure that causes attendance data loss or corruption, creates false attendance results, exposes sensitive biometric or account information, bypasses access control, or makes the core attendance process unavailable.

Major defect: A failure that prevents an important function from operating correctly but does not cause permanent data loss or a security breach. Examples include being unable to activate a session, complete verification, apply a valid manual correction, or access attendance records.

Minor defect: A failure that causes limited inconvenience without preventing the main function from being completed, such as a layout issue, unclear non-critical message, or minor reporting problem.

The system shall not be released with any known unresolved critical defects. Major defects affecting authentication, attendance marking, verification, data integrity, or manual review shall be resolved before production release.

3.4 Performance and Security

The system shall provide acceptable response times during attendance activities and shall protect user accounts, biometric information, location data, attendance records, and verification results from unauthorized access or modification. The following targets shall be evaluated through performance, load, and security testing.

3.4.1 Transaction Response Time

Under normal operating conditions:

- Login, timetable retrieval, attendance-session activation, geofence validation, and QR validation shall have an average response time of no more than 2 seconds.
- Face verification shall normally return a result in less than 5 seconds.
- The lecturer dashboard shall display a completed student verification result within 3 seconds after the backend records it.
- Standard attendance reports shall be displayed within 5 seconds. Exporting reports containing up to 10,000 attendance records shall complete within 20 seconds.

Network delays outside the system's control are not considered when evaluating backend response times.

3.4.2 Throughput and Capacity

The system shall support at least 1,000 eligible students within a single attendance session and at least three concurrently active attendance sessions.

For a session containing up to 1,000 students, the system shall accept attendance submissions throughout the configured check-in window without losing requests, creating duplicate attendance records, or associating results with the wrong student or session.

The system shall support at least 10 attendance-related submissions per second under sustained load and short bursts of at least 20 submissions per second for API request acceptance and lightweight operations, including session access, geofence validation, QR validation, and attendance-status retrieval.

Face-verification requests shall be placed in a controlled processing queue when their arrival rate exceeds the available Face Verification worker capacity. Students shall be shown a Pending status while processing continues. The required number of workers and completed face verifications per second shall be determined through performance testing on the selected deployment infrastructure.

3.4.3 Degraded Operation and Resource Utilisation

When the arrival rate of face-verification requests exceeds the available processing capacity, the system shall accept valid requests into a controlled queue and display their status as Pending. Requests received within the active verification window shall not be rejected solely because

immediate processing capacity is unavailable.

A failure or slowdown in non-critical functions, including notifications, analytics, and report generation, shall not prevent core functions such as authentication, attendance submission, verification processing, or attendance-record storage from operating where technically possible.

When the Face Verification Service is unavailable, the system shall not treat the affected requests as successful. The requests shall be retried according to the configured policy or marked for lecturer review.

Under the tested workload, deployed services should normally operate below 80% sustained CPU and memory utilisation. When utilisation exceeds the configured threshold for a sustained period, the system shall record the condition and generate an operational alert.

Temporary information may be stored in Redis for caching and queued processing, while authoritative attendance records and completed verification results shall be stored in PostgreSQL.

3.4.4 Authentication and Access Security

All protected functions and backend endpoints shall require authentication. Role and permission checks shall be performed by the backend for every protected request.

Passwords shall be stored using an approved adaptive password-hashing algorithm and shall never be stored or transmitted in plain text. The system shall apply secure session expiration, rate limiting, and temporary access restrictions after repeated failed login attempts [2], [3].

Students, lecturers, and administrators shall access only the information and functions authorised for their assigned roles.

3.4.5 Communication and Sensitive Data Protection

All communication between the mobile application, web dashboard, backend services, database connections, and external integrations shall use encrypted communication protocols. Internet-facing API communication shall use HTTPS with TLS 1.2 or higher, while database and internal service connections shall use TLS or another secure encrypted connection when supported [2]-[4].

Sensitive information, including reference face embeddings, temporary face images, location information, authentication credentials, and attendance records, shall be protected against unauthorized access. Face embeddings and other sensitive stored information shall be encrypted at rest where supported [2], [3], [5], [6].

Temporary live images shall be deleted after verification or completion of an authorised review according to the configured retention policy. Application logs shall not contain passwords, authentication tokens, raw face embeddings, QR secrets, or unnecessary biometric information [2], [3], [5], [6].

3.4.6 Attendance Verification Security

Dynamic QR tokens shall be integrity-protected, short-lived, linked to the correct attendance session and QR verification session, and validated using server-controlled time. The system shall reject repeated submissions of the same QR token by the same student within a single QR verification session.

Face verification shall compare the live face only with the authenticated student's reference

embedding. Liveness checks shall be completed before the live image is accepted for verification.

The system shall reject altered requests, expired verification activities, duplicate submissions, replayed tokens, unauthorised session access, and verification results that cannot be validated reliably. A failed or unavailable security-related verification shall not be treated as successful attendance.

3.4.7 Auditability and Security Testing

The system shall maintain protected audit records for login attempts, account changes, biometric reference updates, attendance-session actions, verification results, manual overrides, configuration changes, and administrative activities.

Audit records shall not be modifiable or deletable by students or ordinary lecturers.

The system shall undergo authentication testing, authorisation testing, API security testing, input-validation testing, dependency vulnerability scanning, and testing for unauthorised access before release. The production release shall contain no known unresolved critical security vulnerability [2], [3].

3.5 Supportability

The system shall be designed so that developers and authorised technical personnel can diagnose faults, maintain components, update configurations, deploy new versions, and extend the system without causing unnecessary disruption to unrelated functions.

3.5.1 Coding Standards and Maintainable Code

All system components shall follow documented coding standards, naming conventions, project structures, formatting rules, and error-handling practices appropriate to their technologies.

Source code shall be organised into clearly defined modules with limited responsibilities. Reusable functions and shared libraries shall be used where appropriate to reduce duplicated code.

3.5.2 Technical Documentation

The project shall maintain up-to-date technical documentation covering:

- System architecture and component responsibilities
- Backend API endpoints
- Database schema and migrations
- Environment and configuration variables
- Deployment and maintenance procedures
- Face-verification and attendance-decision workflows

Documentation shall be updated when significant interfaces, configurations, or system behaviour change.

3.5.3 Logging and Fault Diagnosis

The system shall generate structured logs for application errors, service failures, database errors, background-processing failures, and verification-processing problems.

Logs shall include the timestamp, affected service, severity level, request or correlation identifier, and

sufficient diagnostic information to identify the cause of a failure. Sensitive information such as passwords, authentication tokens, raw face embeddings, and unnecessary biometric data shall not be included.

3.6 Design Constraints

The system shall be developed according to the following mandatory architectural, technology, platform, and standards-related constraints.

3.6.1 Architectural Constraint

The backend shall follow a modular service-oriented architecture with clearly defined responsibilities and documented interfaces between components.

The Face Verification Service shall operate as a separately deployable service because its image-processing and model-inference workload may require independent scaling. The core backend shall contain the attendance-management functions, including an Attendance Verification component responsible for coordinating verification activities and determining the final attendance result.

3.6.2 Technology Stack

The system shall be implemented using the below project technology stack:

- Next.js with TypeScript for the lecturer and administrator web dashboard [11]
- React Native with Expo for the student mobile application [12], [13]
- Python FastAPI for the backend APIs [14]
- PostgreSQL as the primary relational database [9]
- Redis for caching, temporary state, rate limiting, queuing, and background-processing support [10]
- OpenCV for backend image preprocessing [15]
- DeepFace with the pretrained ArcFace model for face-embedding extraction [17]
- Google ML Kit Face Detection for mobile-side face detection and basic liveness support [16]

PostgreSQL shall serve as the authoritative data store for permanent and transactional information. Redis shall not be used as the sole permanent storage for attendance records or completed verification results [9], [10].

3.6.3 Standards Compliance Constraint

The web dashboard and online user manual shall follow WCAG 2.2 Level AA accessibility requirements where applicable [1].

The student mobile application shall follow relevant Material Design 3 [7] and Apple Human Interface Guidelines [8] to maintain familiar and accessible behaviour across supported platforms.

The system shall also comply with the security, privacy, interoperability, and institutional standards identified in Section 3.12, Applicable Standards. Where university policies impose stronger requirements than the general design constraints, the institutional requirements shall take precedence.

3.6.4 Client Platform and Hardware Constraint

The student application shall be implemented as a mobile application supporting the approved Android and iOS platforms. The lecturer and administrator interfaces shall be implemented as

browser-based web applications.

The system shall use the student's mobile-device camera and location capabilities for face capture, liveness verification, geofence validation, and QR scanning. The system shall not require specialised classroom hardware such as RFID readers, Bluetooth beacons, UWB devices, biometric scanners, or permanently installed classroom cameras.

3.7 On-line User Documentation and Help System Requirements

3.7.1 Online User Manual

The system shall provide a role-specific online user manual as a static website hosted on Vercel or an equivalent platform [18]. The mobile application and web dashboard shall contain a clearly visible link to the manual.

The manual shall provide instructions for:

- Student account activation, first-time password change, and trial face verification
- Student attendance check-in, liveness verification, face capture, geofence validation, and QR scanning
- Lecturer attendance-session creation, verification activation, monitoring, review, and finalisation
- Administrator account management, academic data synchronisation, classroom and geofence configuration, and face-reference updates

The manual shall also provide troubleshooting guidance for common problems, including failed login attempts, unsuccessful trial face verification, poor image quality, failed liveness checks, disabled location permissions, inaccurate location results, expired QR codes, network failures, and pending attendance reviews.

The documentation shall use clear language, screenshots or illustrations where useful, and separate navigation for each user role.

3.7.2 Documentation Maintenance and System Information

The online user manual shall be updated whenever a significant user-facing function, workflow, or interface changes. It shall display the system version or last updated date so that users can confirm whether the instructions are current.

The system shall provide an About or Help section containing the application version, privacy information, supported-device guidance, and the approved institutional support or contact method.

3.8 Purchased Components

The system shall not require any purchased software component or specialised commercial hardware for its initial implementation and deployment.

The project shall use open-source libraries, freely available development tools, and cloud services that may offer free or paid service plans.

3.9 Interfaces

The system shall provide clearly defined interfaces between its users, client applications, backend components, device capabilities, and authorised external systems.

3.9.1 User Interfaces

The system shall provide a mobile application for students and a browser-based web dashboard for lecturers and administrators. The interfaces shall present only the functions authorised for the authenticated user's role and shall provide clear status, validation, loading, error, and confirmation messages.

Student Mobile Application

The student mobile application shall provide interfaces for authentication, active and upcoming attendance sessions, enrolled courses, attendance history, face and liveness verification, geofence validation, optional QR scanning, notifications, profile management, and online help.

3.9.1.1 Student Login Interface

The student login interface shall contain fields for the university username and password, a password visibility control, a login button, and a password-recovery option. It shall display validation messages for incomplete or incorrect credentials, together with the application version and an authorised-use notice. A self-registration option shall not be provided because student accounts are created by administrators.

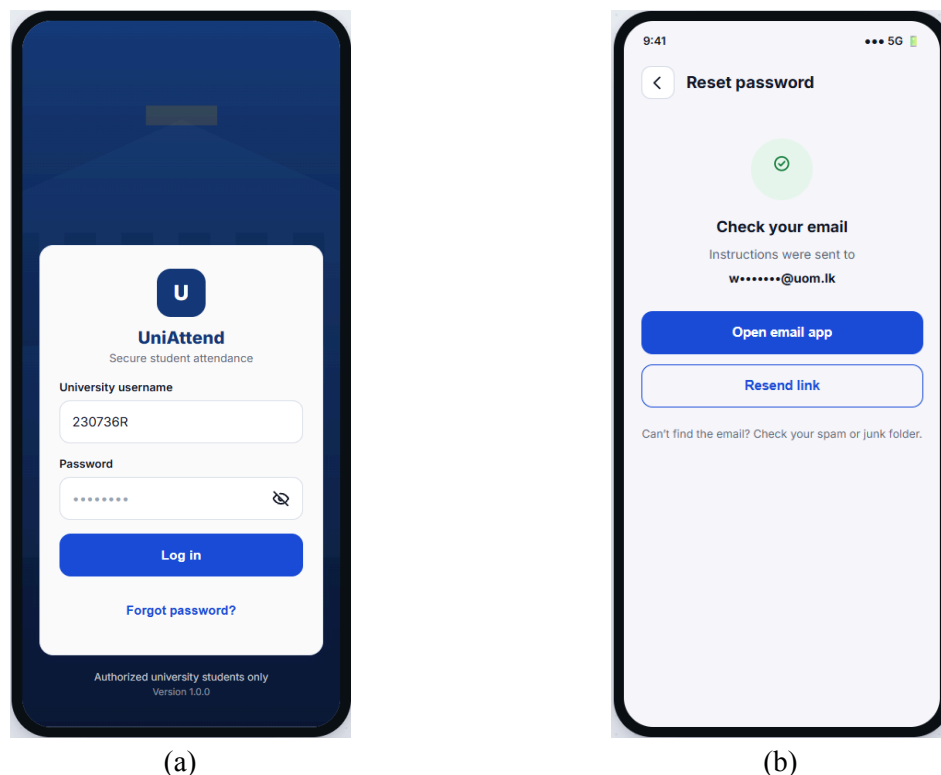


Figure 1. Student authentication interfaces: (a) Student Login Interface (b) Password Reset Confirmation Interface.

3.9.1.2 Student Home Interface

The student home interface shall display any active attendance session together with the course name, session time, classroom, remaining check-in time, and an action to begin attendance verification. It shall also display enrolled courses, attendance percentages, upcoming lectures, notifications, and profile access. The primary navigation menu shall contain Home, Courses, and Notifications.

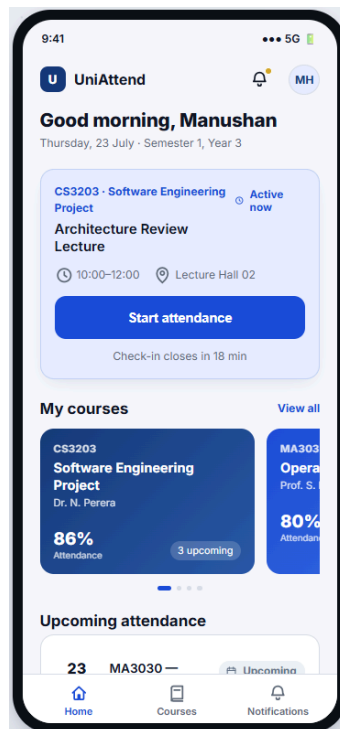


Figure 2. Student Home Interface

3.9.1.3 Course and Attendance Interface

The courses interface shall display the student's enrolled courses with their course code, course name, lecturer, academic period, and attendance percentage. It shall provide a search field and suitable filters, together with separate views for scheduled sessions and attendance records. Session information shall show relevant statuses such as Upcoming, Active, Present, Late, Absent, Pending Review, Cancelled, or Close.

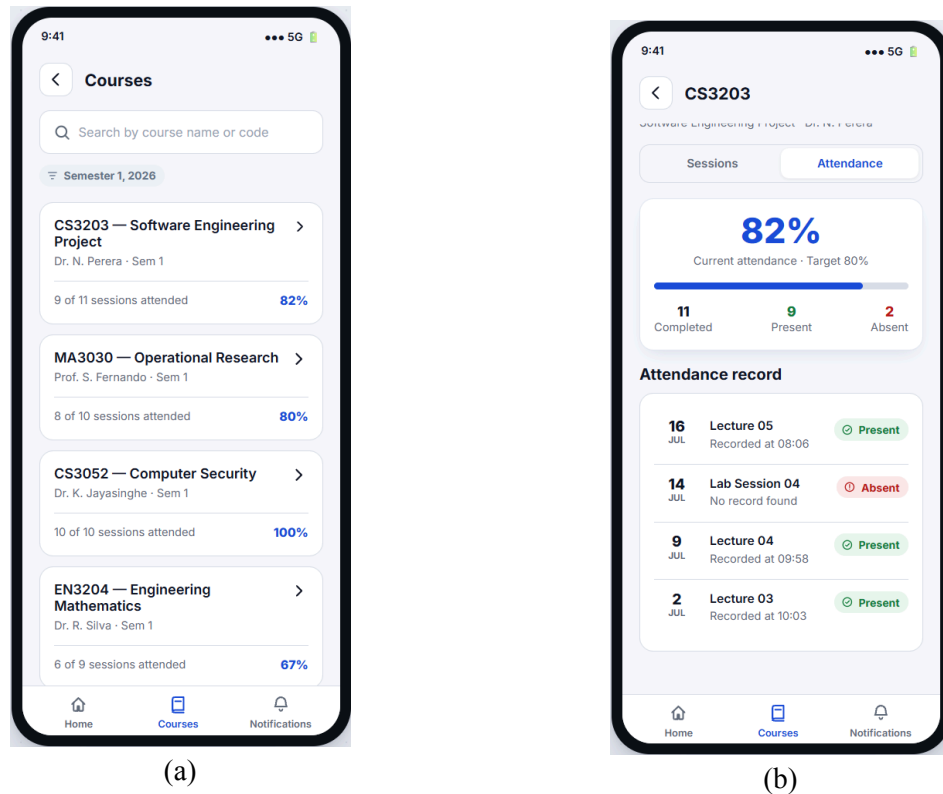


Figure 3. Student course and attendance interfaces: (a) Course List Interface and (b) Course Attendance Record Interface.

3.9.1.4 Attendance Verification Interface

The attendance-verification interface shall display the course, lecturer, classroom, session date and time, remaining verification period, and current attendance status. It shall provide instructions and camera guidance for liveness and face verification, including image-quality, processing, success, failure, and retry states. After successful face verification, the application shall indicate that the initial check-in has been recorded and that further verification may be requested.

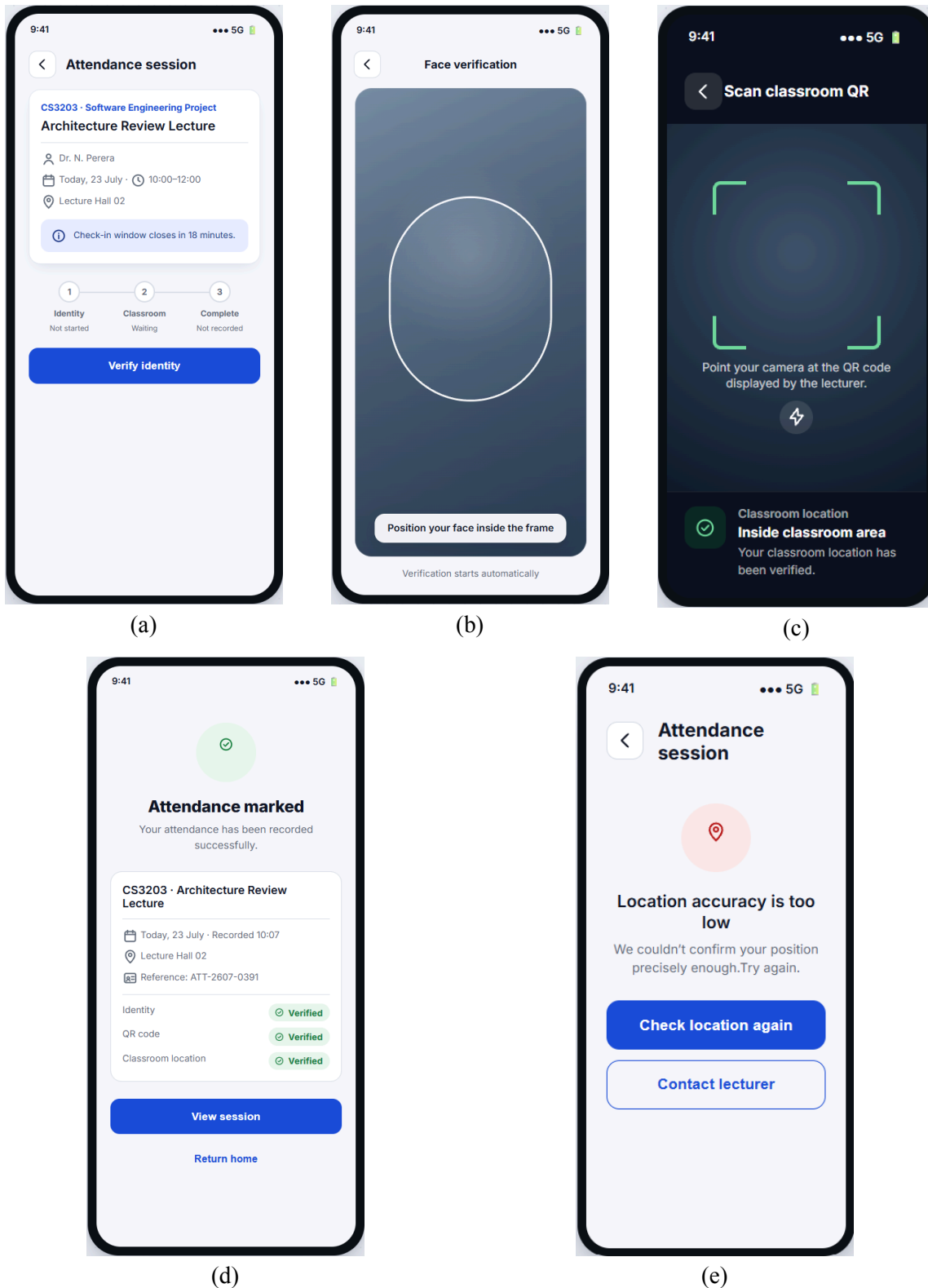


Figure 4. Student attendance verification workflow: (a) Attendance Session Details Interface, (b) Face Verification Interface, (c) Dynamic QR Scanning and Geofence Confirmation Interface, (d) Attendance Confirmation Interface, and (e) Location Accuracy Error and Lecturer Contact Interface.

3.9.1.5 Notifications and Profile Interfaces

The student application shall provide a notification interface for attendance-session and additional-verification alerts. The profile interface shall display the student's university account information and provide access to password change, privacy information, application settings, the online user manual, and logout.

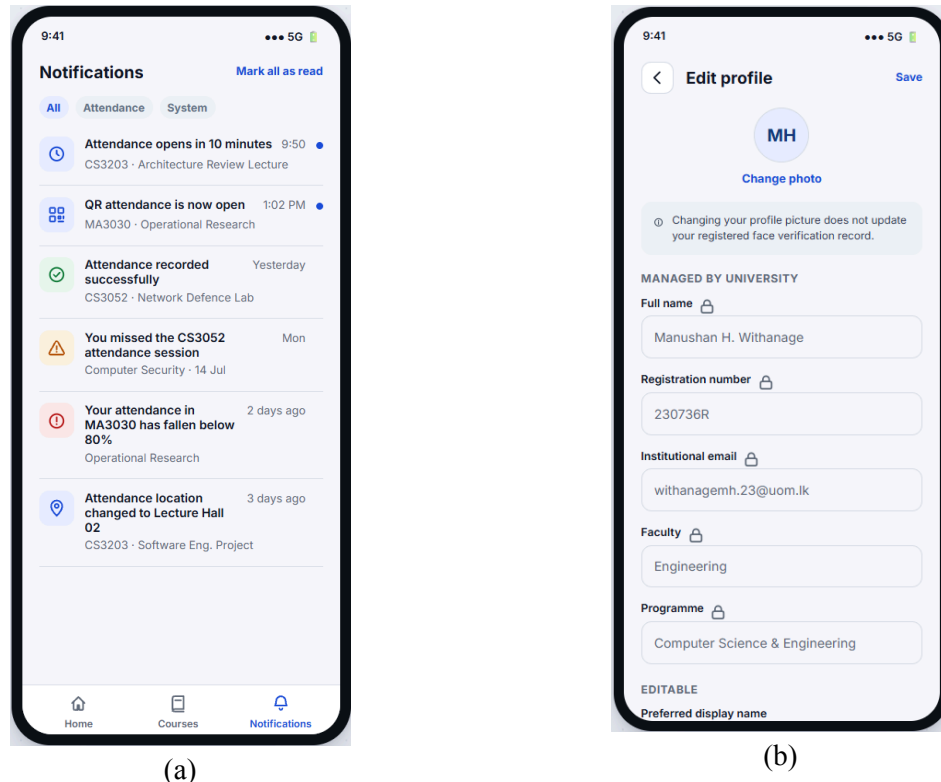


Figure 5. Student account interfaces: (a) Notifications Interface and (b) Profile Management Interface.

Lecturer Web Dashboard

3.9.1.6 Lecturer Login Interface

The lecturer login interface shall contain university username and password fields, a password-visibility control, a login button, and a password-recovery option. It shall display validation, role-restricted access, and authorised-use notices. Lecturer accounts shall be created by administrators, and self-registration shall not be provided.

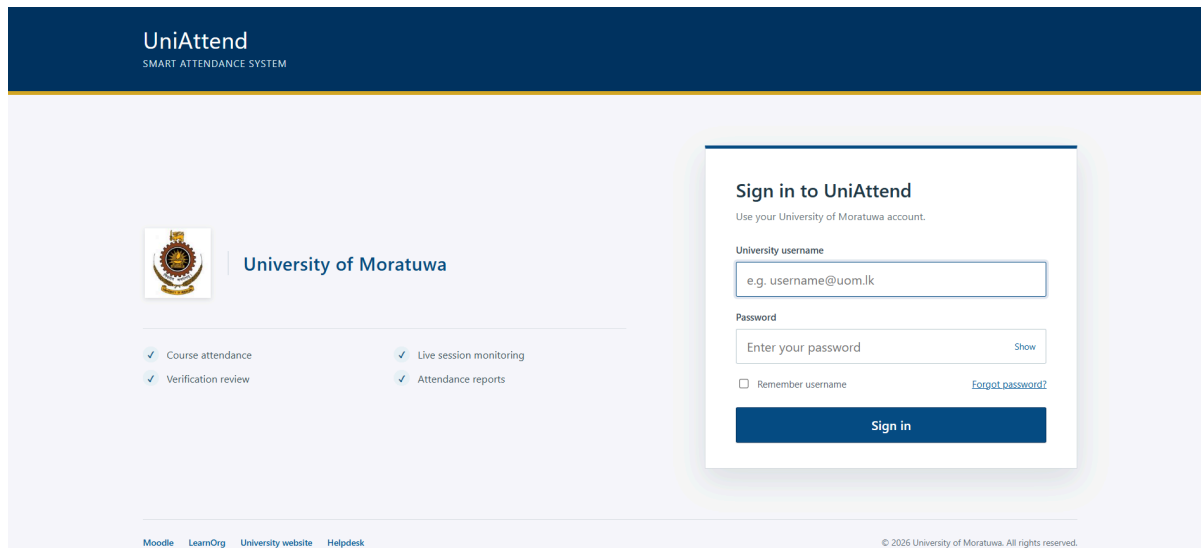


Figure 6. Lecturer and Administrator Login Interface.

3.9.1.7 Lecturer Main Dashboard Interface

The lecturer dashboard shall display active and upcoming sessions, pending verification reviews, attendance summaries, recent alerts, and assigned courses. It shall provide direct access to live-session monitoring and controls for activating and closing optional QR verification sessions.

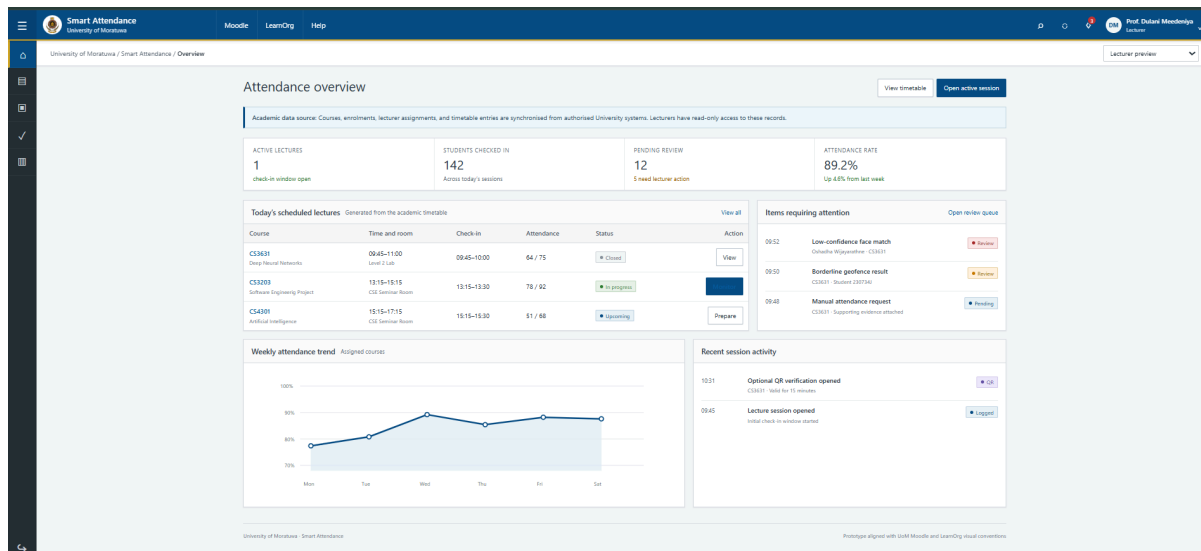


Figure 7. Lecturer Main Dashboard Interface.

3.9.1.8 Lecturer Session and Attendance Interface

The Sessions interface shall display upcoming, active, and completed sessions using searchable and filterable tables. The live-session monitor shall display attendance counts, student verification statuses, check-in times, and active verification events, while allowing the lecturer to start and end optional QR checks and to close or finalise the attendance session.

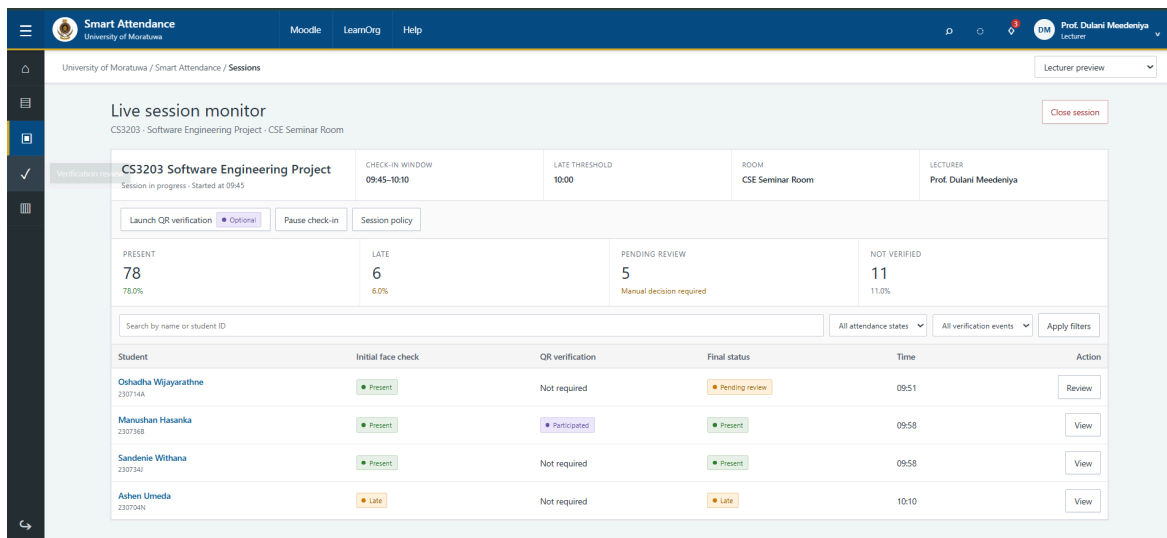


Figure 8. Lecturer Attendance Session Management Interface.

3.9.1.9 Verification Review and Reporting Interface

The Verification Review interface shall display flagged student cases, face-verification results, geofence evidence, optional QR evidence, and authorised review actions. The Reports and Analytics interface shall provide filters, attendance summaries, charts, student risk information, and report-export options.

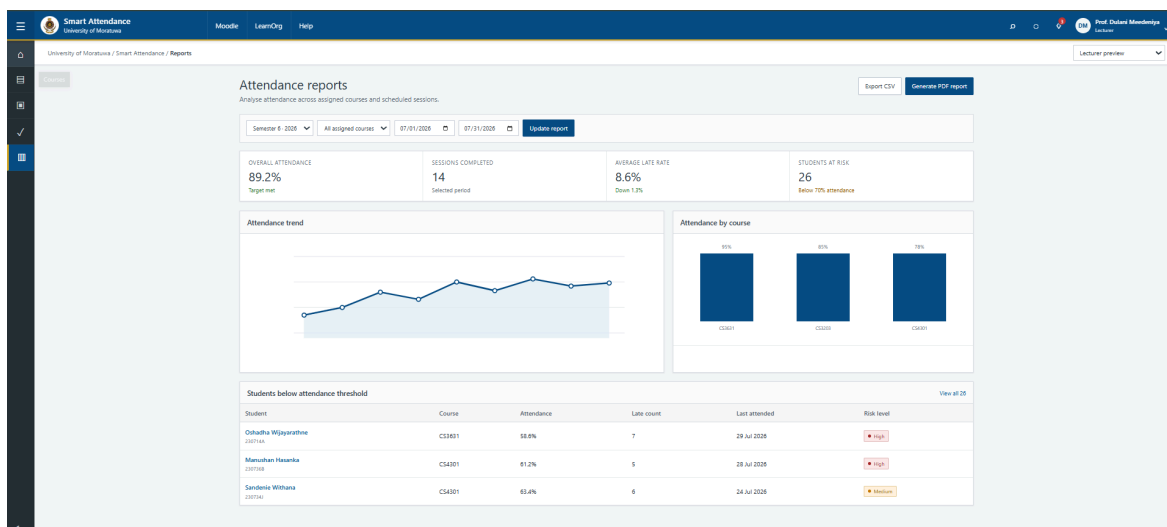


Figure 9. Lecturer Attendance Reporting Interface

Administrator Web Dashboard

The administrator web dashboard shall provide interfaces for managing user accounts and roles, synchronising academic data, configuring classrooms and geofences, maintaining student reference-face profiles, managing institution-wide attendance policies, reviewing audit logs, monitoring system health, and generating institution-level reports. It shall display administrative alerts, pending access or photo-update requests, verification risks, recent configuration changes, and system status information, while restricting these functions to authorised administrators.

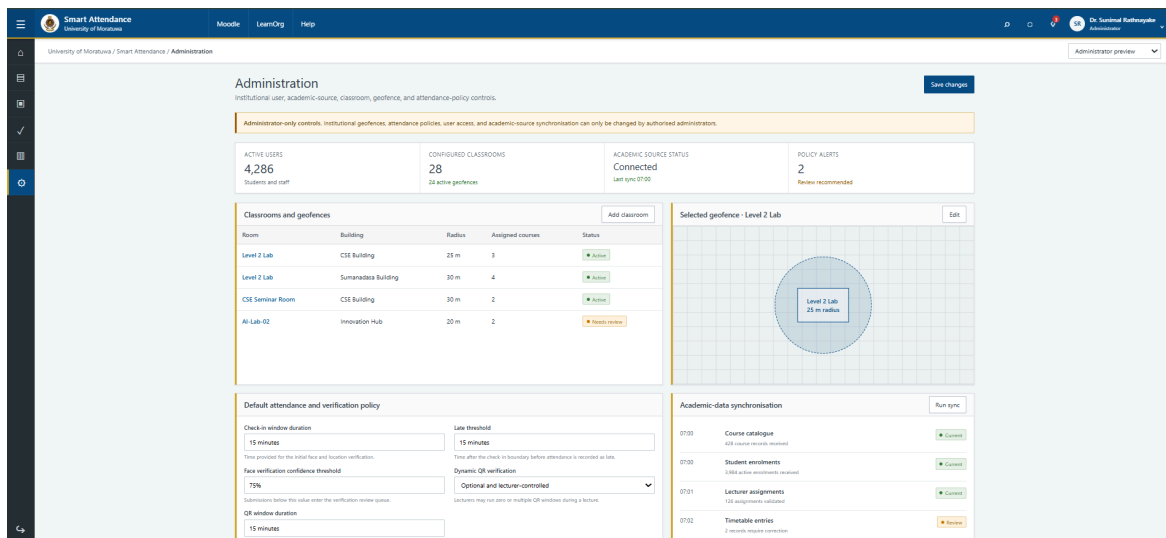


Figure 10. Administrator Overview.

3.9.2 Hardware Interfaces

The student mobile application shall interface with the device's front-facing camera for face capture and liveness verification, camera for QR scanning, GPS and location services for geofence validation, network interface for server communication, and notification capability for attendance alerts. The application shall request the required permissions and display an appropriate error or alternative action when a camera, location service, network connection, or permission is unavailable.

Lecturers and administrators shall access the web dashboard using a laptop or desktop computer with a supported modern web browser and internet connection. The system shall not require direct access to physical hardware addresses or specialised classroom equipment such as RFID readers, biometric scanners, Bluetooth beacons, UWB devices, or permanently installed cameras. Hardware interaction shall be performed through the mobile operating system and browser-provided interfaces.

3.9.3 Software Interfaces

The student mobile application and lecturer/administrator web dashboard shall communicate with the core backend through authenticated REST APIs. The core backend shall communicate with the separately deployable Face Verification Service through authenticated internal APIs for image preprocessing, embedding extraction, and one-to-one face comparison. PostgreSQL shall be used as the authoritative persistent data store, while Redis shall support temporary session state, caching, rate limiting, and queued verification requests [9], [10].

The student application shall use Google ML Kit and mobile operating-system services for on-device face detection and basic liveness checks [16]. The backend shall interface with the university LMS or another authorised academic source through an approved API or validated file-import mechanism to obtain user, course, enrolment, lecturer-assignment, and timetable information. The system shall also interface with a platform push-notification service for attendance and verification alerts. Client applications shall not connect directly to PostgreSQL, Redis, or internal backend services.

3.9.4 Communication Interfaces

The student mobile application and the lecturer/administrator web dashboard shall communicate with the backend over the internet using asynchronous HTTPS requests secured with TLS 1.2 or later through TCP port 443 [4]. REST interfaces shall exchange UTF-8 encoded JSON data, while face

images shall be transmitted using secure multipart or binary requests with file-type and size validation. The lecturer dashboard shall receive live attendance updates through secure WebSocket connections, with HTTPS polling used as a fallback when WebSocket communication is unavailable.

Communication between the core backend, Face Verification Service, PostgreSQL, and Redis shall occur only through authenticated or access-restricted private-network connections using configurable service addresses and ports. Internal databases and services shall not be directly accessible from public client applications. Network failures, timeouts, retries, and duplicate requests shall be handled without creating duplicate attendance or verification records.

3.10 Database Requirements

3.10.1 Core Data Storage

The database shall store the information required for:

- User accounts, roles, and profiles
- Imported academic information, enrolments, and lecturer assignments
- Classrooms, geofence configurations, and timetables
- Attendance sessions and verification windows
- Student reference face embeddings and their update history
- Face, liveness, geofence, and QR verification results
- Student attendance records and check-in times
- Manual attendance updates and audit records

Temporary live face images shall not be treated as permanent database records.

3.10.2 Data Integrity and Relationships

The database shall enforce appropriate primary keys, foreign keys, uniqueness constraints, and required fields to maintain consistent relationships between students, lecturers, courses, attendance sessions, verification attempts, and attendance records.

The database shall ensure that each student has only one authoritative attendance record for a particular attendance session. Duplicate submissions shall not create duplicate final attendance records.

Related database operations shall use transactions where partial completion could create inconsistent data.

3.10.3 Historical Data Preservation and Retrieval

Changes to user accounts, enrolments, lecturer assignments, timetables, classrooms, geofence configurations, or reference face embeddings shall not remove or incorrectly modify historical attendance information.

The database shall support efficient retrieval of attendance records based on student, course offering, attendance session, date, academic period, attendance status, and lecturer. Frequently searched fields shall be indexed, and large results shall support pagination.

3.11 Licensing, Legal, Copyright, and Other Notices

3.11.1 Data Protection and Privacy

The system shall process student identity information, facial data, location data, and attendance records in accordance with applicable data-protection requirements and the policies of the adopting institution [5], [6].

Before biometric and location data are used, students shall be informed about the information collected, the purpose of collection, how it is used, who may access it, and how long it will be retained. Location data shall be collected only during attendance-verification activities and shall not be used for continuous tracking [5], [6].

Temporary live face images shall be removed according to the approved retention policy. Access to biometric, location, and attendance information shall be limited to authorised users and system components [5], [6].

3.11.2 Copyright, Trademarks, and Acceptable Use

The project's original source code, documentation, diagrams, interface assets, and user manual shall contain appropriate copyright and ownership notices. Unauthorised copying, modification, or distribution of protected project materials shall not be permitted.

Third-party names, logos, trademarks, images, and other protected materials shall be used only according to the applicable permissions and usage guidelines.

The system shall provide privacy and acceptable-use notices informing users that they shall not share accounts, impersonate another student, submit another person's facial data, share active QR codes, interfere with attendance verification, or attempt unauthorised access to the system.

3.12 Applicable Standards

3.12.1 Accessibility and User Interface Standards

The lecturer and administrator web dashboard and the online user manual shall conform to WCAG 2.2 Level AA, including the applicable Level A and Level AA requirements for keyboard accessibility, readable content, colour contrast, form labels, error identification, and navigation [1].

The student mobile application shall follow relevant Material Design 3 guidance for Android [7] and Apple Human Interface Guidelines [8] for iOS to provide familiar, consistent, and accessible user interactions across supported devices.

3.12.2 Application Security Standards

The web dashboard, backend APIs, authentication functions, access-control mechanisms, session management, file uploads, logging, and sensitive-data handling shall be designed and tested using the relevant requirements of OWASP ASVS 5.0 [2].

The student mobile application shall also follow relevant controls from the OWASP MASVS, particularly those concerning secure storage, authentication, network communication, platform interaction, and privacy [3].

External and internal communications shall use HTTPS with TLS 1.2 or later, with TLS 1.3 used where supported. TLS 1.3 is defined by RFC 9846 [4].

3.12.3 Data Protection and Institutional Requirements

The collection and processing of facial data, face embeddings, location information, identity information, and attendance records shall comply with the applicable provisions of the Personal Data Protection Act, No. 9 of 2022 [5], as amended by the Personal Data Protection (Amendment) Act, No. 22 of 2025 [6].

The system shall also comply with the student-data, biometric-data, academic-record, information-security, and acceptable-use policies of the adopting university. Where institutional requirements are stronger than the general project requirements, the institutional requirements shall take precedence.

3.12.4 Date and Time Interoperability Standard

Dates and timestamps exchanged between the mobile application, web dashboard, backend services, Face Verification Service, PostgreSQL, Redis, and external academic systems shall use ISO 8601-1:2019 compatible representations [19].

Server-controlled time shall be used when evaluating attendance check-in windows, late thresholds, QR verification windows, QR expiry times, and audit timestamps.

3.12.5 Documentation Standards

The SRS, Software Architecture Document, reports, references, diagrams, and other academic deliverables shall follow the documentation format prescribed by the university templates. References and citations shall use the required IEEE referencing style.

4. Supporting Information

No additional supporting information is included in this SRS. The user-interface illustrations required to clarify the interface requirements are provided in Section 3.9. All requirements are defined in Section 3 of this document.